

Apply filters to SQL queries

Project description

In this project, I used SQL queries to investigate security-related data from login attempts and employee records. I applied filters to retrieve specific information from the `log_in_attempts` and `employees` tables, including failed login attempts, login activity on specific dates, login attempts outside Mexico, and employee machine information for security updates.

Retrieve after hours failed login attempts

To investigate failed login attempts that occurred after business hours, I queried the `log_in_attempts` table and filtered for attempts after 18:00 where the login was unsuccessful.

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_time > '18:00'
-> AND success = FALSE;
```

event_id	username	login_date	login_time	country	ip_address	success
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
18	pwashing	2022-05-11	19:28:50	US	192.168.66.142	0
20	tshah	2022-05-12	18:56:36	MEXICO	192.168.109.50	0

This query uses the `AND` operator to return only records that meet both conditions: the login attempt happened after 18:00 and the login attempt failed.

Retrieve login attempts on specific dates

To investigate suspicious activity on 2022-05-09 and the day before, I queried login attempts from both dates.

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE login_date = '2022-05-09'
-> OR login_date = '2022-05-08';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1
4	dkot	2022-05-08	02:00:39	USA	192.168.178.71	0

This query uses the `OR` operator to return login attempts from either 2022-05-09 or 2022-05-08.

Retrieve login attempts outside of Mexico

To identify login attempts that did not originate in Mexico, I used the `NOT` and `LIKE` operators.

```
MariaDB [organization]> SELECT *
-> FROM log_in_attempts
-> WHERE NOT country LIKE 'MEX%';
```

event_id	username	login_date	login_time	country	ip_address	success
1	jrafael	2022-05-09	04:56:27	CAN	192.168.243.140	1
2	apatel	2022-05-10	20:27:27	CAN	192.168.205.12	0
3	dkot	2022-05-09	06:47:41	USA	192.168.151.162	1

This query excludes countries that begin with `MEX`, which accounts for both `MEX` and `MEXICO`.

Retrieve employees in Marketing

To identify employee machines in the Marketing department located in the East building, I queried the `employees` table.

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Marketing'
-> AND office LIKE 'East%';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1052	a192b174c940	jdarosa	Marketing	East-195
1075	x573y883z772	fbautist	Marketing	East-267

This query uses `AND` to filter for employees who are in Marketing and whose office begins with `East`.

Retrieve employees in Finance or Sales

To identify employees in either the Finance or Sales departments, I used the `OR` operator.

```
MariaDB [organization]> SELECT *
-> FROM employees
-> WHERE department = 'Finance'
-> OR department = 'Sales';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170
1009	NULL	lrodrigu	Sales	South-134
1010	k242l212m542	jlansky	Finance	South-109
1011	l748m120n401	drosas	Sales	South-292
1015	p611q262r945	jsoto	Finance	North-271
1017	r550s824t230	jclark	Finance	North-188
1018	s310t540u653	abellmas	Finance	North-403
1022	w237x430y567	arusso	Finance	West-465
1024	y976z753a267	iuduike	Sales	South-215

This query returns employees from either department so their machines can receive the correct security update.

Retrieve all employees not in IT

To identify employees outside the Information Technology department, I used the `NOT` operator

```
MariaDB [organization]> SELECT *
```

```
-> FROM employees
```

```
-> WHERE NOT department = 'Information Technology';
```

employee_id	device_id	username	department	office
1000	a320b137c219	elarson	Marketing	East-170
1001	b239c825d303	bmoreno	Marketing	Central-276
1002	c116d593e558	tshah	Human Resources	North-434
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366

This query returns all employees who are not in Information Technology, allowing the team to identify which machines still require updates.

Summary

In this project, I used SQL filters to retrieve information from security-related datasets. I applied `AND`, `OR`, `NOT`, and `LIKE` operators to investigate login attempts and identify employee machines requiring security updates. This project helped demonstrate how SQL can support cybersecurity investigations and administrative security tasks.